

MacroSnaps: Data and Stories Audit

Last revised March 12, 2026. Reflects commodity price automation (`fetch_market_data.py`) and commodity story automation (`update_commodity_stories.py`) completed March 12, 2026.

Helicopter View

Data / Content Type	Source	Script	Frequency
Macro metrics (GDP, CPI, Unemployment, Budget, CA, Policy Rate)	Google Sheet	<code>sync_sheet.py</code>	When consensus forecasts change (infrequent)
Market metrics (Stock YTD, 10Y Yield, Yield Curve, FX pair)	Yahoo Finance / FRED	<code>fetch_market_data.py</code>	Daily
Data-void metrics (Equity Vol, Corp Spread, Sov CDS, FX Vol)	No free source	N/A	Displayed as "not available"
Historical chart arrays	FRED / Yahoo Finance	<code>refetch_historical.py</code>	Manually, when needed
Commodity prices	Yahoo Finance (continuous futures)	<code>fetch_market_data.py</code>	Daily (automated)
Per-metric stories (168 blocks)	Claude API (Sonnet)	<code>update_stories.py</code>	After each data fetch, for metrics that moved past threshold
Country headline stories (108 blocks)	Claude API (Haiku + harvest)	<code>update_headlines.py</code>	Daily
Global stories (9 blocks)	Claude API (Sonnet + web search)	<code>update_headlines.py</code>	Daily
Commodity stories (27 blocks)	Claude API (Haiku)	<code>update_commodity_stories.py</code>	Daily (automated, threshold-gated)
metricBriefs	Manual (written once)	None	Rarely
fxRegime descriptions	Manual (written once)	None	Rarely

Part 1: Data Series

1A. Macro Metrics (6 per country)

These are annual consensus forecasts. Source of truth is Ralph's Google Sheet. Written into `data.json` by `sync_sheet.py`. Updated infrequently, only when consensus forecasts change. Tier: **structural** (GDP Growth, Budget Deficit, Current Account) or **weekly** (Inflation, Unemployment, Policy Rate).

Metric	Update frequency	Source	Script	Notes
GDP Growth	Structural (rarely)	Google Sheet	<code>sync_sheet.py</code>	Year-end 2026 consensus forecast
Inflation (CPI)	Weekly	Google Sheet	<code>sync_sheet.py</code>	Year-end 2026 consensus forecast
Unemployment	Weekly	Google Sheet	<code>sync_sheet.py</code>	Year-end 2026 consensus forecast
Budget Deficit	Structural (rarely)	Google Sheet	<code>sync_sheet.py</code>	Year-end 2026 consensus forecast. Historical arrays empty for 9 of 12 countries.
Current Account	Structural (rarely)	Google Sheet	<code>sync_sheet.py</code>	% of GDP (converted March 10, 2026 using IMF WEO denominators). Historical arrays complete.
Policy Rate	Weekly	Google Sheet	<code>sync_sheet.py</code>	Year-end 2026 forecast, not current rate

Current values by country:

Country	GDP Growth	Inflation	Unemployment	Budget Deficit	Current Account	Policy Rate
USA	+2.2%	2.3%	4.2%	-7.5% GDP	-3.8% GDP	3.25%
CAN	+1.7%	2%	6.5%	-1.5% GDP	-0.5% GDP	2.25%
GBR	+1.5%	2.1%	5.0%	-4.2% GDP	-2.5% GDP	3.5%
JPN	+0.8%	1.8%	2.5%	-1.2% GDP	+4.8% GDP	1.25%
DEU	+1.2%	2.2%	3.8%	-2% GDP	+4.5% GDP	2.50%
FRA	+1.4%	2%	7.7%	-4.8% GDP	+1% GDP	2.50%
ITA	+1%	2.3%	5.6%	-3.2% GDP	+0.8% GDP	2.50%
CHN	+4.7%	2%	5.1%	-6.8% GDP	+2.5% GDP	2.75%
IND	+6.3%	4.2%	4.7%	-7.2% GDP	-1.2% GDP	6%
ZAF	+1.5%	3.5%	32.5%	-4.8% GDP	-1.4% GDP	6.25%
BRA	+1.7%	4.2%	5.7%	-0.8% GDP	-2.6% GDP	12%
RUS	+0.9%	4.7%	2.5%	-2.5% GDP	+2% GDP	12%

1B. Market Metrics (8 per country)

Fetched daily by `fetch_market_data.py` from Yahoo Finance and FRED. Tier: **daily**.

Metric	Source	Fetch method	Notes
Stock Market YTD	Yahoo Finance	YTD % change from Jan 1 close	Fails for RUS (IMOEX.ME delisted)
Equity Vol	Yahoo Finance	Implied vol index; 30-day realized vol as fallback	Data-void: displayed as "not available"
10Y Bond Yield	FRED	Daily series per country	Fails for CHN and BRA (no FRED series)
Yield Curve	FRED (derived)	10Y minus short rate	Fails for CHN and BRA; fixed for USA/DEU/FRA/JPN in March 2026
Corp Spread	FRED	ICE BofA IG/HY OAS series	Data-void: displayed as "not available". No RUS series configured.
Sov CDS	Derived proxy	Local 10Y minus UST (EM only)	Data-void: displayed as "not available"
FX pair	Yahoo Finance	Varies by country (see table below)	Succeeds for all countries
FX Vol	Yahoo Finance (derived)	30-day realized vol from daily FX returns	Data-void: displayed as "not available"

FX pairs by country:

Country	Pair	Current value	Last updated
USA	USD/DXY	99.2	2026-03-11
CAN	CAD/USD	0.74	2026-03-11
GBR	GBP/USD	1.34	2026-03-11
JPN	USD/JPY	158.8	2026-03-11
DEU	EUR/USD	1.1575	2026-03-11
FRA	EUR/USD	1.1575	2026-03-11
ITA	EUR/USD	1.1575	2026-03-11

Country	Pair	Current value	Last updated
CHN	USD/CNY	6.86	2026-03-11
IND	USD/INR	92.15	2026-03-11
ZAF	USD/ZAR	16.40	2026-03-11
BRA	USD/BRL	5.15	2026-03-11
RUS	USD/RUB	79.1	2026-03-11

Current market metric values by country:

Country	Stock YTD	Equity Vol	10Y Yield	Yield Curve	Corp Spread	Sov CDS	FX Vol
USA	-1.4%	~25	4.12%	+52bps	84bps	42bps	5.8%
CAN	+3.7%	~20	3.40%	+121bps	84bps	45bps	5.8%
GBR	+4.0%	~15	4.45%	+74bps	84bps	32bps	6.8%
JPN	+6.2%	~32	2.24%	+112bps	84bps	28bps	8.5%
DEU	-3.6%	~20	2.81%	+81bps	84bps	20bps	6.3%
FRA	-2.0%	~16	3.53%	+153bps	84bps	38bps	6.3%
ITA	-1.3%	~22	3.49%	+149bps	84bps	70bps	6.3%
CHN	+2.7%	~13	2.40%	+35bps	162bps	72bps	3.2%
IND	-9.8%	~21	6.73%	+123bps	162bps	261bps	7.2%
ZAF	+1.1%	~29	8.62%	+187bps	162bps	450bps	17.2%
BRA	+14.8%	~22	13.80%	-140bps	162bps	180bps	10.5%
RUS	+5.0%	~32	13.5%	-140bps	380bps	320bps	15.0%

Known permanent data gaps (not bugs):

- RUS: Stock Market YTD and Equity Vol always fail (IMOEX.ME delisted on Yahoo)
- CHN and BRA: 10Y Bond Yield, Yield Curve, and Sov CDS always fail (no FRED series)
- RUS: Corp Spread not configured
- Equity Vol, Corp Spread, Sov CDS, FX Vol: all four are data-void metrics with no reliable free daily source. All displayed as "not available" in UI.

1C. Historical Chart Data (`\_frozen\_historical`)

Populated by `refetch_historical.py` from FRED and Yahoo Finance. Updated manually, not daily. Never touch by hand.
Each metric has 7 data points: actuals for 2020-2025, plus the 2026F forecast as the final point.

Population status by country:

Country	Charts populated	Known gaps
USA	14/14	None
CAN	14/14	Budget Deficit array empty
GBR	14/14	Budget Deficit array empty
JPN	14/14	None
DEU	14/14	Budget Deficit array empty
FRA	14/14	Budget Deficit array empty
ITA	14/14	None
CHN	11/14	Unemployment, 10Y Bond Yield, Yield Curve: no free source. Budget Deficit array empty.
IND	11/14	Unemployment, 10Y Bond Yield, Yield Curve: no free source. Budget Deficit array empty.

Country	Charts populated	Known gaps
ZAF	12/14	Unemployment: no monthly source. Budget Deficit array empty.
BRA	12/14	10Y Bond Yield, Yield Curve: no free source. Budget Deficit array empty.
RUS	10/14	GDP Growth, Unemployment, historical data discontinued post-2022 sanctions. Budget Deficit array empty.

Known data quality flags:

- IND 2025 GDP Growth (16.77) and ZAF 2025 GDP Growth (6.76) appear to reflect nominal USD growth from FRED rather than real % growth. Pending investigation.
- Budget Deficit historical arrays are intentionally empty for 9 countries (CAN, GBR, DEU, FRA, CHN, IND, ZAF, BRA, RUS). A single forecast point with no history would be meaningless as a chart.
- Running `refetch_historical.py` overwrites Current Account arrays with raw dollar values, undoing the % of GDP conversion. Must re-apply the CA conversion after any refetch run.

1D. Commodities (9 items)

Source: CME, ICE, COMEX, CBOT, Yahoo Finance. Updated manually as part of daily content session. Tier: **daily** for price and spark array.

Each commodity has: price, change (YoY % from Yahoo Finance), spark (rolling 12-point array), `annual` (long-run annual price history), `storyWrittenAtPrice` (price at time of last story write, used for drift detection), and `storyUpdatedDate`.

Commodity	Symbol	Category	Unit	Current price	Last updated
WTI Crude	CL	Energy	\$/bbl	94.19	Mar 12, 2026
Brent Crude	BZ	Energy	\$/bbl	95.30	Mar 12, 2026
Natural Gas	NG	Energy	\$/mmBtu	3.23	Mar 12, 2026
Gold	GC	Metals	\$/oz	5,185	Mar 12, 2026
Silver	SI	Metals	\$/oz	87.23	Mar 12, 2026
Copper	HG	Metals	\$/lb	5.87	Mar 12, 2026
Wheat	ZW	Agriculture	cents/bu	604	Mar 12, 2026
Corn	ZC	Agriculture	cents/bu	465.75	Mar 12, 2026
Soybeans	ZS	Agriculture	cents/bu	1,229	Mar 12, 2026

How commodity data is updated: `fetch_market_data.py` pulls price, YoY change, spark array, and `asOf` for all 9 commodities daily via Yahoo Finance continuous futures tickers (CL=F, BZ=F, NG=F, GC=F, SI=F, HG=F, ZW=F, ZC=F, ZS=F). No manual edits needed. The `annual` array is long-run history and updated rarely.

Part 2: Stories and Headlines

2A. Per-Metric Stories (168 total)

12 countries x 14 metrics x 3 audience levels (beginner, moderate, expert) = 168 story blocks.

Each story lives inside its metric entry in `data.json` as a story object. Stories appear in the metric tooltip between the value and the chart.

How they are created: `update_stories.py` diffs `data.json` against the last git commit, identifies metrics that moved past a configurable threshold, calls the Claude API (claude-sonnet-4-20250514), and rewrites affected stories at all three levels in one pass.

When to run: After every `sync_sheet.py` --apply and every `fetch_market_data.py` run. Part of the daily bash ritual (Step 5).

Stories for the 4 data-void metrics (Equity Vol, Corp Spread, Sov CDS, FX Vol) are not maintained going forward as those metrics display as "not available."

Completion status: All 168 stories are complete as of March 11, 2026.

Story length targets:

- Beginner: 2-3 sentences. No jargon. Plain English explanation of what the number means for everyday life.
- Moderate: 3-4 sentences. One piece of context (historical comparison, regional comparison, or causal driver).
- Expert: 4-5 sentences. Specific numbers, a directional signal, one forward-looking implication.

Known value/story drift (as of March 12, 2026):

The following metrics have live values that have moved since their stories were last written. These need a story rewrite session.

Country	Metric	Current value	Story references
USA	Stock Market YTD	-1.4%	~+2%
USA	Equity Vol (VIX)	~25	~16
USA	10Y Bond Yield	4.12%	4.28%
USA	Yield Curve	+52bps	+8bps
USA	USD/DXY	99.2	104.2
USA	FX Vol	5.8%	8.5%
CAN	Stock Market YTD	+3.7%	6.8%
CAN	10Y Bond Yield	3.40%	(check story)
CAN	Yield Curve	+121bps	+30bps
CAN	CAD/USD	0.74	(check story)
CAN	FX Vol	5.2%	7.2%
GBR	Stock Market YTD	+4.0%	(check story)
GBR	10Y Bond Yield	4.45%	(check story)
GBR	Yield Curve	+74bps	(check story)
GBR	GBP/USD	1.34	(check story)
JPN	Equity Vol	~32	(check story)
JPN	10Y Bond Yield	2.24%	(check story)
JPN	Yield Curve	+112bps	(check story)
JPN	USD/JPY	158.8	(check story)
DEU	Stock Market YTD	-3.6%	(check story)
DEU	10Y Bond Yield	2.81%	(check story)
DEU	Yield Curve	+81bps	(check story)
DEU	EUR/USD	1.1575	(check story)
FRA	Stock Market YTD	-2.0%	(check story)
FRA	10Y Bond Yield	3.53%	(check story)
FRA	Yield Curve	+153bps	(check story)
ITA	Stock Market YTD	-1.3%	(check story)
ITA	10Y Bond Yield	3.49%	(check story)
ITA	Yield Curve	+149bps	(check story)
CHN	Stock Market YTD	+2.7%	(check story)
CHN	USD/CNY	6.86	(check story)

Country	Metric	Current value	Story references
IND	Stock Market YTD	-9.8%	(check story)
IND	10Y Bond Yield	6.73%	(check story)
IND	Yield Curve	+123bps	(check story)
IND	Sov CDS	261bps	(check story)
IND	USD/INR	92.15	(check story)
ZAF	Stock Market YTD	+1.1%	(check story)
ZAF	Equity Vol	~29	(check story)
ZAF	10Y Bond Yield	8.62%	(check story)
ZAF	Yield Curve	+187bps	(check story)
ZAF	Sov CDS	450bps	(check story)
ZAF	USD/ZAR	16.40	(check story)
BRA	Stock Market YTD	+14.8%	(check story)
BRA	USD/BRL	5.15	(check story)
RUS	USD/RUB	79.1	(check story)

2B. Country-Level Headline Stories (12 countries)

Each country has a stories block with 3 bullet points per level (beginner, moderate, expert). These are the top-of-card narrative bullets visible on each country panel.

How they are created: `update_headlines.py` runs a two-phase pipeline:

1. Phase 1: A single Sonnet + web search "harvest" call pulls recent macro data for all 12 countries (capped at 2 search turns, 1500 max tokens).
2. Phase 2: 3 Haiku batches (4 countries each) write story bullets with recent data leading and forecast values as background context only.

The script writes a draft to `stories_draft_YYYY-MM-DD.json`. Review is done in `headline_review.html` (browser tool). After approval, apply with `python3 update_headlines.py --apply stories_approved_YYYY-MM-DD.json`.

Update frequency: Daily, as part of the daily bash ritual (Step 6).

Current status: All 12 countries complete. Last full run: March 12, 2026 (first successful run with two-phase architecture, 13/13 calls successful).

Story format per country: 3 bullets at each of 3 levels (9 blocks per country, 108 blocks total).

2C. Global Stories (3 slots)

Three top-of-page story cards that frame the global macro picture for the day. Each slot has 3 levels (beginner, moderate, expert) totaling 9 text blocks.

Three-act structure (fixed):

- Slot 1 (Today's Story): dominant macro event of the day
- Slot 2 (Biggest Movers): which markets, currencies, or economies are reacting and how
- Slot 3 (The Connection): what ties slots 1 and 2 together, the "so what" for the global picture

How they are created: Same `update_headlines.py` pipeline. Global runs first (before country stories) to avoid Sonnet rate limit exhaustion from the harvest call. Uses Sonnet + web search, 5000 max tokens.

Update frequency: Daily.

Current global stories (as of March 12, 2026):

- Slot 1: Iran Crisis Shuts Key Oil Route
 - Slot 2: (Biggest Movers)
 - Slot 3: (The Connection)
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2D. Commodity Stories (9 commodities)

Each commodity has a `story` object with `beginner`, `moderate`, and `expert` keys (each as a `{"text": "..."} object`). These describe the commodity market context and outlook.

How they are created: `update_commodity_stories.py` runs daily as part of the bash ritual (Step 5, immediately after `fetch_market_data.py`). It compares each commodity's current price against `storyWrittenAtPrice` stored in `data.json`. Any commodity that has moved past its threshold gets new stories written at all three levels via a single Haiku API call. Changes are applied directly to `data.json` with no review step.

Thresholds:

- Natural Gas: 10%
- All other commodities: 5%

Update frequency: Daily (automated). On quiet days the script exits in under a second with no API call made.

Last bootstrapped: March 12, 2026. All 9 `storyWrittenAtPrice` values set.

Story length targets:

- Beginner: 2-3 sentences. What this commodity is and why its price matters to ordinary people.
 - Moderate: 3-4 sentences. One driver of the current price level and one downstream effect.
 - Expert: 4-5 sentences. Specific price, directional trend, supply/demand driver, one forward implication.
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2E. metricBriefs (short summaries)

Each country has a `metricBriefs` object with a short explanatory sentence for each of its 14 metrics. These appear in the tooltip as a plain-English "what is this metric" blurb.

Update frequency: Rarely. These are definitional and do not change with data.

Status: Complete for all 12 countries.

2F. fxRegime Descriptions

Each country has a `fxRegime` object describing its foreign exchange policy at three levels (beginner, moderate, expert).

Update frequency: Rarely. Structural information that only changes if a country changes its FX policy regime.

Status: Complete for all 12 countries.

Part 3: Summary - How Everything Fits Together

Data / content type	Source	Update script	Frequency
Macro metrics (GDP, CPI, Unemployment, Budget, CA, Policy Rate)	Google Sheet	<code>sync_sheet.py</code>	When consensus forecasts change (infrequent)
Market metrics (Stock YTD, 10Y Yield, Yield Curve, FX pair)	Yahoo Finance / FRED	<code>fetch_market_data.py</code>	Daily
Data-void metrics (Equity Vol, Corp Spread, Sov CDS, FX Vol)	No free source	N/A	Displayed as "not available"
Historical chart arrays	FRED / Yahoo Finance	<code>refetch_historical.py</code>	Manually, when needed

Data / content type	Source	Update script	Frequency
Commodity prices	Yahoo Finance (continuous futures)	fetch_market_data.py	Daily (automated)
Per-metric stories (168 blocks)	Claude API (Sonnet)	update_stories.py	After each data fetch, for metrics that moved past threshold
Country headline stories (108 blocks)	Claude API (Haiku + harvest)	update_headlines.py	Daily
Global stories (9 blocks)	Claude API (Sonnet + web search)	update_headlines.py	Daily
Commodity stories (27 blocks)	Claude API (Haiku)	update_commodity_stories.py	Daily (automated, threshold-gated)
metricBriefs	Manual (written once)	None	Rarely
fxRegime descriptions	Manual (written once)	None	Rarely